



Full Dynamics?



A partir das equações governantes do estado média da atmosfera.

$$\frac{\partial U}{\partial t} + \frac{1}{a \cos^2 \theta} \left(U \frac{\partial U}{\partial \lambda} + V \cos \theta \frac{\partial U}{\partial \theta} \right) + \dot{\eta} \frac{\partial U}{\partial \eta} - fV + \frac{1}{a} \left(\frac{\partial \Phi}{\partial \lambda} + R_d T_v \frac{\partial \ln p}{\partial \lambda} \right) = F_u \quad (1)$$

$$\frac{\partial V}{\partial t} + \frac{1}{a \cos^2 \theta} \left(U \frac{\partial V}{\partial \lambda} + V \cos \theta \frac{\partial V}{\partial \theta} \right) + \dot{\eta} \frac{\partial V}{\partial \eta} + fU + \frac{\cos \theta}{a} \left(\frac{\partial \Phi}{\partial \theta} + R_d T_v \frac{\partial \ln p}{\partial \theta} \right) + \frac{\sin \theta}{a \cos^2 \theta} (U^2 + V^2) = F_v \quad (2)$$

$$\frac{\partial T}{\partial t} + \frac{1}{a \cos^2 \theta} \left(U \frac{\partial T}{\partial \lambda} + V \cos \theta \frac{\partial T}{\partial \theta} \right) + \dot{\eta} \frac{\partial T}{\partial \eta} - \frac{\kappa T_v \omega}{(1 + (\delta - 1)q)p} = F_T \quad (3)$$

$$\frac{\partial q}{\partial t} + \frac{1}{a \cos^2 \theta} \left(U \frac{\partial q}{\partial \lambda} + V \cos \theta \frac{\partial q}{\partial \theta} \right) + \dot{\eta} \frac{\partial q}{\partial \eta} = F_q \quad (4)$$

$\theta = \text{latitude}$
 $\lambda = \text{Longitude}$
 $U = \text{Velocidade Zonal}$

$$\frac{\partial \ln p_s}{\partial t} + \frac{1}{p_s} \int_0^1 \nabla \cdot \left(\mathbf{v}_H \frac{\partial p}{\partial \eta} \right) d\eta = 0 \quad (5)$$

$$\frac{\partial \phi}{\partial \eta} + \frac{R_d T_v}{p} \frac{\partial p}{\partial \eta} = 0 \quad (6)$$



Atmosphere Numerical Modeling Group

1) Implemente este modelo baseado nas **Equações do Estado Média Da atmosfera**

https://github.com/pkubota/pgmet_2025/tree/main/MET-348-3/scm_dynamics_2025

2) Integre o modelo para o caso do furacão Catarina e compare os campos de **velocidade do vento**, **Temperatura** e **Umidade Especifica** com os dados do Era5?

Link dos dados do furacão Catarina

https://ftp.cptec.inpe.br/pesquisa/bam/paulo.kubota/externo/MET-348-3/scm_pgmet_2025.tar.gz

https://ftp.cptec.inpe.br/pesquisa/bam/paulo.kubota/externo/MET-348-3/scm_dynamics_2025/dataout/

